Yuhao Xie

LLM Inference | KV Cache Quantization | GPU Kernel Optimization

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Computer Engineering PhD student focused on efficient LLM inference, with an **EMNLP 2026 main-conference first-author paper** and experience integrating CUDA quantization kernels into an LLM inference engine.

EDUCATION

University of Central Florida | PhD, Computer Engineering — Aug. 2023 – Aug. 2027 (expected)

University of Florida | MS, Electrical & Computer Engineering — Aug. 2021 – May 2023

Hangzhou Dianzi University | BS, Electrical & Computer Engineering — Sept. 2017 – June 2021

PUBLICATION

Yuhao Xie, Mingjie Lin.

Beyond: Better-than-Full-Precision KV Caches via Learnable Non-Uniform Quantization as Implicit Regularization.

EMNLP 2026, main conference. [Paper] [Code] [Project]

LLM RESEARCH & ENGINEERING

Hithink RoyalFlush Information Network — May – Aug. 2024

Research Intern, Large Language Models — Zhejiang, China

- Developed CUDA quantization/dequantization kernels, reducing execution time by **90% versus a PyTorch implementation**; integrated them into the company's LLM inference engine with **FlashAttention and paged KV-cache support**.
- Implemented **2-bit KV-cache quantization** with separate grouping along the sequence and head dimensions for keys and values; achieved **50%** lower GPU memory use and **8**× throughput for the company's 7B financial LLMs on a single NVIDIA A10.
- Tuned quantization group size and residual length on internal benchmarks to retain **99.3% of the original model's accuracy**.

SELECTED SYSTEMS PROJECTS

Fused Attention Decoding with Quantized KV Caches

- Fused **on-the-fly KV dequantization** with attention scoring, online softmax, and value accumulation; reconstructed packed K/V in registers using groupwise quantization parameters, avoiding full-precision KV materialization in HBM.
- Achieved a **40% inference speedup for Llama 2 7B** on a single **NVIDIA H100** versus a Hugging Face implementation.

TileCadence — Shared-Memory Scheduling for FlashAttention-2 [Code]

- Built a **constraint solver** to reassign freed shared memory to probability tiles, output fragments, and LSE. Compiled **CUDA/CuTe** kernels for Ampere and Blackwell and ranked schedules by resource use and measured latency.
- Measured **1.11–1.13**× attention speedups over default FlashAttention-2 across 13 Qwen3-4B prefill shapes (8K–32K) on **DGX Spark**, with bitwise-identical outputs and LSE.

ADDITIONAL ENGINEERING EXPERIENCE

Rtelligent Technology | Software Engineer Intern — Jan. – May 2021

- Developed ARM Cortex-M firmware in C++; reduced G-code processing time by **20%** and used Xilinx FPGA raster-image parallelism to increase laser engraving speed by **30%**.

TECHNICAL SKILLS

Programming & tools: Python, C/C++, CUDA, OpenMP, Verilog, Git, Docker.
ML systems: PyTorch, Transformers, vLLM, FlashAttention, KV-cache quantization, speculative decoding.

TEACHING, SERVICE & HONORS

UCF: Instructor: Aug. – Dec. 2026: Digital Systems (EEE3342C) and Computer Organization (EEL3801C); Aug. 2024 – May 2025: Computer Organization Lab; **Teaching Assistant:** Aug. – Dec. 2026: Hardware Security and Trusted Circuit Design (EEE4346C); May – Aug. 2025: Junior Design.
Reviewer: ICLR 2027.
ORCGS Doctoral Fellowship, UCF (2023–2027).
Zhejiang provincial collegiate programming competition, team third prize (2019); academic scholarships, Hangzhou Dianzi University (2017–2020).